



**Universidade de Brasília
Faculdade de Tecnologia
Departamento de Engenharia Mecânica**

**Gated Data-Space Inversion for
Reservoir Prediction**

Thiago Tomás de Paula

DISSERTAÇÃO DE MESTRADO
PROGRAMA DE PÓS-GRADUAÇÃO EM CIÊNCIAS MECÂNICAS

Brasília
2026

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Agregação de Modelos Adaptativa
Aplicada a Otimização de Reservatórios

Thiago Tomás de Paula

Dissertação de Mestrado submetida ao Programa de Pós-Graduação em Ciências Mecânicas como requisito parcial para obtenção do grau de Mestre.

Orientador: Prof. Dr. Eugênio Libório Feitosa Fortaleza

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Trabalho aprovado. Brasília, 15 de julho de 2026:

Prof. Dr. Eugênio Libório Feitosa Fortaleza
UnB/FT/ENM
Orientador

Examinador interno

Examinador interno

Abstract

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Keywords: ; ; .

Resumo

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Palavras-chave: ; ; .

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1 Introduction

1.1 Chapter outline

Aims and structure

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1.2 Contextualization

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1.3 Motivation

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

1.4 Contributions of this work

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1.5 Objectives of this dissertation

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1.6 Dissertation outline

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1.7 Chapter summary

Key takeaways

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2 Literature Review

2.1 Chapter outline

Aims and structure

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2.2 Chapter summary

Key takeaways

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3 Theory framework

3.1 Chapter outline

Aims and structure

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

3.2 Elements of reservoir engineering

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

3.2.1 Reservoir formation

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

3.2.2 Reservoir simulation

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3.3 Elements of machine learning

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

3.3.1 Unsupervised vs. Supervised vs. Reinforcement Learning

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

3.3.2 Regression and Classification

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3.3.3 Ensemble Methods

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

3.4 Data-Space Inversion

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3.5 Chapter summary

Key takeaways

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4 Methods

4.1 Chapter outline

Aims and structure

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4.2 Reservoir case studies

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

4.2.1 Egg ensemble

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

4.2.2 Olympus ensemble

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

4.2.3 Real field

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4.3 Chapter summary

Key takeaways

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5 Results

5.1 Chapter outline

Aims and structure

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5.2 Chapter summary

Key takeaways

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6 Discussion

6.1 Chapter outline

Aims and structure

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6.2 Chapter summary

Key takeaways

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7 Conclusion

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7.1 Future works

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([Schaum, 1956](#))



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